INCH-POUND

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MILITARY SPECIFICATION

INSTRUMENTS, ELECTRICAL-MEASURING, HYDROGEN-GAS-MONITORING, NONPORTABLE (SUBMARINE USE)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the production and testing of nonportable hydrogen-gas monitoring systems for use on submarines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Benificial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard; Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-I-631	- Insulation, Electrical, Synthetic Resin Composition, Non-Rigid.
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-K-3926	- Knobs, Control (For Use With Electronic, Communications, and Allied Equipment).
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-F-15160	- Fuses: Instrument, Power and Telephone.
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-A-17196	- Alarm Switchboards, Alarm Panels, Remote Sentry Alarm Switch, Signal Units, and Special Alarm Connection Box, Naval Shipboard.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140	- Lumber and Plywood Fire-Retardant Treated.
MIL-D-28803/1	- Display, Optoelectronic, Segmented Readout, Backlighted, Style II (Light Emitting Diodes), RFI Shielded, Moisture Sealed, High Impact Shock, Type R01
MIL-H-46855	- Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-T-55164	- Terminal Boards, Molded, Barrier Screw and Stud Types, and Associated Accessories, General Specification for.
MIL-D-87157	- Displays, Diode, Light Emitting, Solid State, General Specification for.

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STANDARDS

MILITARY

MIL-STD-12	- Abbreviations for Use on Drawings, and in Specifications, Standards, and in Technical Documents.
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-740-1	- Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2	- Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1132	- Switches and Associated Hardware, Selection and Use of.
MIL-STD-1346	- Relays Selection and Application.
MIL-STD-1353	- Electrical Connectors, Plug-In Sockets and Associated Hardware, Selection and Use of.
MIL-STD-1399	- Interface Standard for Shipboard Systems
Section 300	- Electric Power, Alternating Current (Metric).
DOD-STD-1399	- Interface Standard for Shipboard Systems
Section 301	- Ship Motion and Attitude (Metric).
MIL-STD-2000	- Standard Requirements for Soldered Electrical and Electronic Assemblies

HANDBOOKS

MIL-HDBK-241	- Design Guide for Electromagnetic Interference EMI Reduction in Powers Supplies.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA, 19111-5094).

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2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publication Officer, 222 South Riverside Plaza, Room 1106, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Material. Materials forming a part of the finished product shall conform to the requirements of this specification and shall perform their intended function. The metal parts shall be made of a corrosion-resisting material, or of a material treated in a manner to render it resistant to corrosion.

3.2.1 Recovered materials. Equipment, material, and articles incorporated in the products covered by this specification may be fabricated using materials produced from recovered materials to the maximum extent

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practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Exclusion of mercury. The hydrogen monitoring system shall contain no metallic mercury or mercury compounds and shall be free from mercury contamination. During the manufacturing process, tests, and inspections, the hydrogen monitoring system shall not have come in direct contact with mercury or any of its compounds nor with any mercury-containing devices employing only a single boundary of containment. A single boundary of containment is not backed by a second seal or barrier to prevent contamination in the event of a rupture of the primary seal or barrier. Mercury contamination of the supplies shall be cause for rejection of the material.

3.2.3 Exception to mercury exclusion. If the inclusion of metallic mercury or mercury compounds is required as a functional part of the material furnished under this contract, the contractor shall obtain written approval from NAVSEA before proceeding. The contractor's request shall explain in detail the requirements for mercury, identify specifically the parts to contain mercury, and explain the method of protection against mercury escape.

3.3 Monitoring system. The hydrogen monitoring system shall continuously and simultaneously monitor and indicate the concentration of hydrogen in the battery compartment exhaust ventilation ductwork on submarines. The monitoring system shall consist of two remote sensors, four remote indicators, and a controller. The monitoring system shall be provided with visible and audible alarm and malfunction status indicators.

3.3.1 Electrical power system requirements. The monitoring system electrical power system requirements shall be compatible with type I power interface characteristics as specified in MIL-STD-1399, Section 300A. The power supply shall be in accordance with the guidelines of MIL-HDBK-241. Nominal electrical power input and frequency shall be 115 volts, 60 hertz (Hz), single-phase. Electrical stability shall be as specified in 3.5.1.1.

3.3.2 Monitoring system performance requirements.

3.3.2.1 Range of measurement. The monitoring system shall safely measure the hydrogen content of atmospheres in the range of 0 to 4 percent hydrogen (0-4% H_2), by volume (hereinafter expressed only as percent hydrogen).

3.3.2.2 Accuracy. The accuracy of the monitoring system (including stability, conformity, hysteresis, dead band and repeatability) shall be within plus or minus 0.2 percent hydrogen ($\pm$ 0.2% H_2) when exposed to environmental conditions specified in 4.3.

3.3.2.3 Detection level. The monitoring system shall have a minimum detection level of 0.1 percent hydrogen (0.1% H_2).

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3.3.2.4 Operating and non-operating temperature ranges. The monitoring system shall operate within specified accuracy within a temperature range of 0 to 50 degrees Celsius (°C) (32 to 122 degrees Farenheit (°F)) when exposed to pressure and humidity conditions specified in 4.3. The monitoring system shall also conform to specified accuracy and performance requirements after being stored under non-operating conditions as specified in MIL-E-16400 for a sheltered controlled environment.

3.3.2.5 Pressure range. The monitoring system shall operate (within specified accuracy) within an ambient pressure range of 24 to 36 inches of mercury (24-36 in-Hg) when exposed to temperature and humidity conditions specified in 4.3. The monitoring system shall withstand ambient pressures of 10 to 30 pounds per square inch absolute (10-30 psia) without damage or permanent deformation.

3.3.2.6 Humidity. The monitoring system shall operate (within specified accuracy) when exposed to humidity conditions in accordance with MIL-E-16400 and temperature and pressure conditions specified in 4.3.

3.3.2.7 Velocity effects. The monitoring system operations shall not be affected when the sensors are installed in the battery compartment exhaust ventilation ductwork (see 6.2) and subjected to velocity conditions ranging from 0 to 3,000 feet per minute (fpm).

3.3.2.8 Repeatability/reproducibility. Monitoring system repeatability/reproducibility shall be within plus or minus 0.2 percent hydrogen ($\pm$ 0.2% H_2).

3.3.2.9 Response time. The monitoring system shall respond (within specified accuracy) within 30 seconds to reach 90 percent (90%) of the certified value of an applied hydrogen-air or hydrogen-nitrogen mixture.

3.3.2.10 Recovery time. The monitoring system shall return (within specified accuracy) to 90 percent (90%) of its normal ambient reading within 30 seconds after an applied hydrogen-air or hydrogen-nitrogen mixture has been removed.

3.3.2.11 Display range. The controller and remote indicator meter display ranges shall be in accordance with the specified range of measurement (see 3.3.2.1).

3.3.2.12 Accelerated life. The monitoring system shall operate (within specified accuracy) without mechanical or electrical damage during and after subjection to an accelerated life test in accordance with MIL-E-16400.

3.3.2.13 Zero drift. Drift of the zero reading of the controller and remote indicator meters, when operated under steady-state conditions after zero and span adjustment, shall be within plus or minus 0.2 percent hydrogen ($\pm$ 0.2% H_2) during and after subjection to the accelerated life test specified in 3.3.2.12.

3.3.2.14 Drift of span calibration. Drift of the span calibration of the controller and remote indicator meters, when operated under steady-state conditions after zero and span adjustment, shall be within plus or minus 0.2

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percent hydrogen (± 0.2% H_2) during and after subjection to the accelerated life test specified in 3.3.2.12.

3.3.3 Sensor assembly.

3.3.3.1 Type. The monitoring system shall be provided with thermal conductivity or combustion (catalytic) type sensors that are uniquely specific to hydrogen and meet the range-of-measurement requirement specified in 3.3.2.1.

3.3.3.2 Enclosure. Each sensor shall be mounted in its own explosionproof enclosure classified for use in class 1, division 1, group B hazardous locations as specified in article 500 of NFPA 70. Each enclosure shall permit the introduction of sample and calibration gases and the connection of system cables. Each enclosure shall be treated or coated to resist corrosion. Each enclosure shall be suitable for mounting in the battery compartment exhaust ventilation ductwork (see 6.2) and shall not exceed 5-3/4 inches by 6 inches by 7 inches in height, width, and depth, respectively.

3.3.3.3 Calibration port. A calibration port shall be incorporated in the sensor enclosure. The calibration port shall facilitate the connection and introduction of pure air or a certified calibration gas.

3.3.3.4 Sensor output signal. Sensors shall generate a continuous electronic output signal that corresponds to the hydrogen concentration at the sensor. The output signal shall be compatible with the controller scaling circuitry. The output signal shall not be affected by sensor orientation.

3.3.3.5 Terminals. Terminals shall be in accordance with MIL-T-55164. The mounting and marking of terminals shall be in accordance with the appropriate sections of MIL-E-16400.

3.3.3.6 Pressure loss. The sensor shall not cause a pressure loss greater than 0.1 inch of water (0.1 in w.g.) when installed in the battery compartment exhaust ventilation ductwork (see 6.2) and subjected to velocity conditions specified in 4.3.1.

3.3.3.7 Leakage. The sensor enclosure shall be provided with a means to prevent the transmission of air, gas, and moisture between the sensor enclosure and the battery compartment exhaust ventilation ductwork (see 6.2) when the sensor enclosure is mounted to the ductwork and the ductwork is subjected to internal pressures of up to 2.0 pounds per square inch gauge (lb/in^2g).

3.3.4 Controller assembly.

3.3.4.1 Mounting. The necessary instruments, switches, controls, relays and other parts shall be mounted in a single enclosure in such a manner that normally replaceable parts are readily accessible and capable of being removed without special tools. Special tools are defined as those tools not listed in the Federal Supply Catalog.

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3.3.4.2 Arrangement. The necessary metering instruments and controls shall be conveniently arranged on the front of the enclosure. The arrangement of the controls shall be such that normal operating adjustments may be made without opening the enclosure.

3.3.4.3 Analyzing circuit. A separate input circuit shall be furnished for each remote sensor installation.

3.3.4.4 Controller circuitry. The electronic circuitry within the controller, with the exception of interconnecting wiring and conductors, shall be incorporated into self-contained modular units. Each module shall consist of integrated circuits, printed circuit boards, hardwired devices, and discrete plug-in components integrated into a rugged mechanical package. Modules shall be suitable for repeated change-outs and shall be provided with built-in protection to prevent damage due to shorted or open circuits. Major system functions shall be divided among individual modules in such a manner as to expedite trouble-shooting and maintenance. The functional breakdown of the equipment modules shall be as specified in 3.4.6. Microelectronic devices shall be selected and applied in accordance with requirement 64 of MIL-STD-454.

3.3.4.5 Controller output signal. The controller shall generate continuous electronic output signals, corresponding to each sensor input signal, of 4 to 20 milliamps (4-20 mA) which shall correspond to zero to full scale controller and remote indicator meter readings. Controller output signal tolerance shall be within plus or minus 0.65 milliamps (± 0.65 mA) of the actual value.

3.3.4.6 Enclosure. The controller enclosure shall be in accordance with the requirements of MIL-STD-108. The enclosure shall be of sturdy, splashproof construction, and shall be treated or coated to resist corrosion. The enclosure shall be suitable for bulkhead, rack, and panel mounting, and shall not exceed 16-1/2 inches by 16-3/8 inches by 9-5/8 inches in length, width, and depth, respectively. External electrical terminations shall be provided by connectors in accordance with MIL-E-16400. Enclosure materials and components shall be in accordance with MIL-E-16400. The enclosure assembly shall contain a separate meter for each sensor input circuit. A drawout drawer card cage shall be provided to allow the removal of either of the installed circuit modules as a single unit (see 3.3.4.4 and 3.4.6). A single system backplane (motherboard) shall also be provided for the wireless interconnection of interior modules. Sensor and power interconnection terminals shall be mounted inside the enclosure (see 3.3.4.8).

3.3.4.7 Access. Electrical connections between modules and backplane (motherboard) shall be provided by means of rugged quick disconnect connectors. Mechanical guides and fittings shall be provided to ensure that the cage can be inserted into the controller enclosure with the system modules installed beforehand. Installation or removal of individual modules shall be possible with the cage inserted in the enclosure. Removal of the cage shall provide unobstructed access to the system interconnection terminals (see 3.3.4.8). Mechanical keying shall prevent improper insertion of modules into the system backplane.

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3.3.4.8 Terminals. Terminals, protected by a cover that can be removed by the use of captive screws, shall be provided for the connection of the ship's power, sensors, and remote indicators. Terminals shall meet the requirements of MIL-T-55164. The mounting and marking of terminals shall be in accordance with the appropriate sections of MIL-E-16400. Terminals shall be located at the bottom rear of the enclosure and shall be available for unobstructed access from the front of the enclosure when the card cage is removed (see 3.3.4.7). Terminals shall be internally wired to electrical connectors that shall provide complete interconnection with the system backplane (motherboard) when the card is inserted.

3.3.4.9 Switch. A single "off" and "on" switch shall be provided to control each sensor input circuit. Switches and associated hardware shall be selected and applied in accordance with MIL-STD-1132.

3.3.5 Remote indicator assembly.

3.3.5.1 Housing. The remote indicator housing shall be in accordance with the requirements of MIL-STD-108 for splashproof type enclosures. The housing shall be such that all normally replaceable parts are readily accessible and all normal adjustments may be made without opening the enclosure. The housing shall be suitable for panel mounting, and treated or coated to resist corrosion. The housing shall not exceed 4-1/2 inches by 4-1/2 inches by 4-1/2 inches in height, width, and depth, respectively.

3.3.5.2 Terminals. Terminals shall be in accordance with MIL-T-55164. The mounting and marking of terminals shall be in accordance with the appropriate sections of MIL-E-16400.

3.3.6 Marking and identification.

3.3.6.1 Controller identification. Each controller shall have an identification plate, type A, B, C, or H of MIL-P-15024 and MIL-P-15024/5 fastened to the front, containing the following data:

(a) Name of equipment (Hydrogen Monitoring System).
(b) Manufacturer's name, city, and state.
(c) Applicable manufacturer's drawing number (Front View)
(d) Applicable NAVSEA drawing number (Front View)
(e) Manufacturer's model number and serial number.
(f) National stock number.
(g) Voltage, current, and frequency of power to be applied to set.
(h) Contract number and year manufactured.

3.3.6.2 Sensor indicator identification. Sensors shall be provided with identification plates, type A, B, C, or H of MIL-P-15024 and MIL-P-15024/5 containing the following data:

(a) Manufacturer's serial number.
(b) Year manufactured.
(c) Cell type (analysis and reference).

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3.3.6.3 Component identification. The meters, switches, "Power On" lights, and other components on the front of the controller and remote indicators shall be labeled with identification plates-type A, B, C, or H of MIL-P-15024 and MIL-P-15024/5. For example, "Percent Hydrogen," "Power on when Lit," "Power" (under on-off switch), "Alarm," "Malfunction," "Reset," "Test" should be identified, as applicable. Abbreviations, where required due to space limitations, shall be in accordance with MIL-STD-12.

3.3.6.4 Internal wiring, component, and terminal marking. Internal monitoring system wiring shall be marked by stamping on the wire terminals. Where terminals are too small to stamp, wire markers of synthetic resin tubing, type F, grade A of MIL-I-631 or fiber tags may be used. Markings shall agree with the designations shown on the wiring diagram. It is not necessary to mark terminal studs on instruments or control switches, but wiring diagrams shall clearly indicate the studs or contacts to which the marked wires must be connected. Circuit components such as capacitors, resistors, chokes, and so forth shall have markings adjacent to them on the printed circuit card, stamped or otherwise indelibly marked, identifying them with the symbols used in the wiring diagram. Terminal blocks shall have marking strips permanently marked at each terminal with numbers (that is, 1 to 12, and so forth) which agree with the corresponding numbers on the wiring diagram. Terminal blocks shall be numbered TB1, TB2, and so forth.

3.3.7 Meters. Analog/digital type bargraph meters with digital readouts shall be provided for the controller and each remote indicator. Meters shall provide incremental analog/digital displays over the specified range of measurement (see 3.3.2.1). Analog/digital displays shall be scaled in increments of 0.1 percent hydrogen (0.1% H_2) and red-marked from 1.5 percent hydrogen (1.5% H_2) through the upper limit of the specified range of measurement (see 3.3.2.1). Digital readouts shall be in conformance with MIL-D-28803/1 and MIL-D-87157, quality level B. Digital readouts shall provide a display resolution equivalent to 0.01 percent hydrogen (0.01% H_2). Remote indicator meter inputs shall be compatible with the controller output signals (see 3.3.4.5). The controller shall provide operating power to the remote indicator meters.

3.3.8 Status indicators and contacts. A separate visual alarm indicator and a separate malfunction indicator shall be provided within the controller for each sensor input circuit. Separate visual alarm and malfunction indicators shall also be provided for each remote indicator. Parallel audible alarm contacts shall be provided from each circuit for alarm and malfunction conditions. Audible alarm contacts shall be shunted by a 7000 ohm supervisory resistor in accordance with MIL-A-17196. Contacts shall be form (c) type, 120 volts alternating current (Vac), 5 ampere ac rating.

3.3.8.1 Alarm indicators and contacts. Visual alarm indicators and audible alarm contacts shall be set at the factory to respond to conditions where the hydrogen concentration in the battery compartment exhaust ventilation duct increases to a level of 1.5 percent (1.5% H_2). Alarm settings shall be adjustable for a range of 0 to 4 percent hydrogen (0-4% H_2). When an alarm condition is detected, applicable visual alarm indicators shall light up and flash until the reset is engaged.

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3.3.8.2 Malfunction indicator and contacts. A single visual malfunction indicator and audible malfunction contacts shall be provided for each sensor input circuit to respond to conditions where low input power, sensor failure, severed cable, circuit failure, or current failure is experienced. A single pair of relay contacts shall be provided for each sensor input circuit for remote malfunction indication. When a malfunction condition is detected, applicable malfunction indicators shall light up and flash until the reset is engaged.

3.3.8.3 Reset. Manual reset modes shall be provided for the alarm and malfunction indicators and contacts. Separate manual switches shall be provided for each sensor input circuit at the controller and at each remote indicator for resetting the respective alarm indicator and contacts. A manual switch shall also be provided at the controller and at each remote indicator for resetting the malfunction indicator and contacts. Engaging the reset during an alarm or malfunction state shall result in deactivation of the respective audible contacts and cause the respective flashing visual indicators to go to a steady-on state. Visual alarm indicators shall remain in a steady-on state until the reset is engaged with the hydrogen concentration at the affected sensor reduced to a level below the alarm set point (see 3.3.8.1). The visual alarm indicators shall then be turned off. Visual malfunction indicators shall remain in a steady-on state until the reset is engaged and the respective malfunction has been alleviated. The visual malfunction indicators shall then be turned off.

3.3.9 System calibration. A means shall be provided for the manual in-place calibration of the monitoring system when the sensors are subjected to velocities of 315 to 700 fpm. Automatic in-situ calibration shall be permitted as a supplement to manual calibration. A means shall be provided for manually initiating the automatic calibration cycle. The automatic calibration cycle shall not be capable of being initiated automatically. Automatic calibration shall not occur on both sensors simultaneously.

3.3.9.1 Zero adjustment. The monitoring system shall be provided with a means for adjusting each controller meter and associated remote indicator meter to zero when the respective sensor is exposed to a certified zero air. The readout accuracy shall be as specified in 3.3.2.2.

3.3.9.2 Span adjustment. The monitoring system shall be provided with a means for adjusting the span of each controller meter and associated remote indicator meter to full scale when the respective sensor is exposed to a certified hydrogen-nitrogen mixture equal to the upper limit of the specified range of measurement (see 3.3.2.1). The readout accuracy shall be as specified in 3.3.2.2.

3.4 Electrical requirements.

3.4.1 Self-diagnostic capabilities. The monitoring system shall be provided with a means to perform internal monitoring of each circuit to check for shorts, faulty components, or other problems that could prevent proper system response.

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3.4.2 Time delay. The monitoring system shall be provided with a means to temporarily deactivate alarm and malfunction contacts or circuits to prevent false indications at start-up, after power interruption, or during sensor replacement. A visual indicator shall be provided on the controller to indicate when the time delay has expired.

3.4.3 Test mode. A test feature shall be incorporated in the monitoring system to enable the system to be manually tested to verify that the controller, remote indicators, and sensors are operational and to check the system calibration (see 3.3.9 and 3.4.6.4) in the active and passive test modes. Activation of the test circuit shall not result in the energizing of the alarm or malfunction indicator contacts.

3.4.4 Electromagnetic emission and susceptibility. The monitoring system shall meet the requirements of MIL-STD-1399, Section 300A and demonstrate compliance in accordance with the general requirements of MIL-STD-461, part 1, and emission and susceptibility requirements and limits CE01, CE03, CS01, CS06, RE01, RE02, RS01, RS02, and RS03 of MIL-STD-461, part 6, (see 4.7.25).

3.4.5 Circuit boards and components. Electrical parts, mechanical parts, processes, and materials shall be selected and applied in accordance with requirement 17 of MIL-STD-454.

3.4.6 Modules. Modules shall be provided for signal conditioning, readout and control, power supply, system test and calibration, alarm relay, and central processor.

3.4.6.1 Signal conditioner modules. Signal conditioner modules shall house the electronic circuitry for amplifying and monitoring signals generated by each of the remote sensors. The signal conditioner modules may also contain the electrical circuitry for activating alarm and malfunction indicators and contacts (see 3.4.6.6) in accordance with 3.3.8.1 and 3.3.8.2.

3.4.6.2 Readout and control modules. The readout and control modules shall house the electrical circuitry and operator control devices required for hydrogen concentration display, alarm and malfunction status indication and message display, and alarm set point display.

3.4.6.3 Power supply module. The power supply module shall provide a means for generating the supply voltages required for optimum system operating conditions. The power supply module shall operate from a single external electrical power system feed having a nominal electrical power input and frequency as specified in 3.3.1. Power supply outputs shall be distributed to the other system modules through backplane power bus connectors, without loose wiring or jumpers.

3.4.6.4 Controller and remote indicator test and calibration module. The controller and remote indicator test and calibration module shall provide a means for generating a signal that simulates the sensor response characteristics. The signal shall be adjustable over a range equivalent to the full scale range of the monitoring system. The adjustment resolution shall allow determination of system performance.

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3.4.6.5 Alarm relay module. The alarm relay module shall contain switching devices for interfacing with external audible signals conforming to MIL-A-17196. The switching device outputs shall be interconnected to the system terminals through a system backplane (motherboard). The alarm relay module shall contain switching devices for audible alarm and malfunction indication.

3.4.6.6 Central processor module. For microprocessor-based systems, the central processor module shall contain the processor, memory, and interface hardware components required to form the interface system between the signal conditioning modules and readout and control modules. The control logic for the activation of the alarm and malfunction indicators and contacts may reside within the central module as an alternative to the signal conditioner modules (see 3.4.6.1).

3.4.7 Isolated circuits. Sensor input circuits shall be such that the failure of any of the components, sensors, or connecting leads remote indicator within either sensor input circuit shall not interfere with the operation or performance of the other sensor input circuit. The failure of normally replaceable parts, such as pilot lamps, shall not cause damage to other monitoring system components. Malfunction indicators and contacts, however, shall be activated as applicable.

3.4.8 Connectors. Cable connectors shall be selected and applied in accordance with MIL-STD-1353.

3.4.9 Signal amplification. Intermediate signal amplification (see 6.2) shall be provided for conditions that do not allow a stable analog or digital output signal (see 3.3.3.4 and 3.3.4.5) to be maintained between each sensor and the controller, and the controller and each remote indicator.

3.4.10 Relays. Relays shall be selected and applied in accordance with MIL-STD-1346.

3.4.11 Rheostats. Knobs for control rheostats shall be equipped with locking devices or disconnecting clutches in accordance with MIL-K-3926 to prevent accidental change of circuit adjustment.

3.4.12 Fuses. Fuses shall be in accordance with MIL-F-15160. Fuses shall be selected so that the overload blowing characteristics and short circuit interrupting capacity specified in MIL-F-15160 match the overload protection requirements of the equipment and wiring being protected, and the short circuit capacity of the supply circuit.

3.4.13 Burn-in test. The monitoring system shall meet specified performance requirements following 24 hours of continuous operation in accordance with 4.7.26.

3.4.14 Insulation resistance. The insulation resistance shall meet the requirements of MIL-STD-1399, Section 300A and shall not be less than 10 megohms at specified environmental service conditions (see 4.3) when subjected to 500 volts direct current (Vdc) superimposed on the ac power supply (see 4.7.27).

3.5 Environmental performance requirements.

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3.5.1 Stability.

3.5.1.1 Electrical stability. Power line transients and spikes with magnitudes, deviation, repetition rates, and decay characteristics as specified in MIL-STD-1399, Section 300A shall not degrade system performance, cause damage to the monitoring system, or cause any observable permanent change in calibration upon return to steady-state conditions.

3.5.1.2 Thermal stability. Transient changes of plus or minus 10°C (± 10°C) in the steady-state ambient temperature at the controller, remote indicators, or in the environment of the sensors shall not result in a temporary deviation greater than 0.2 percent hydrogen (0.2% H_2) from the calibration value, cause an observable permanent change in calibration upon return to steady-state conditions, or degrade system performance.

3.5.1.3 Pressure stability. Transient and spike changes of plus or minus 2 inches mercury (± 2 in-Hg) in the steady-state ambient pressure shall not result in a temporary deviation greater than 0.2 percent hydrogen (0.2% H_2) from the calibration value, cause an observable permanent change in calibration upon return to steady-state conditions, or degrade system performance.

3.5.2 Shock. The monitoring system shall operate within the specified performance limits and withstand the grade A, class I, type A high impact shock test for lightweight equipment in accordance with MIL-S-901 (see 4.7.31).

3.5.3 Vibration. The monitoring system shall meet the type I environmental vibration test acceptability requirements of MIL-STD-167-1 for a maximum exciting frequency of 25 hertz (Hz) (see 4.7.32).

3.5.4 Noise.

3.5.4.1 Airborne noise. Monitoring system airborne noise levels shall not exceed the requirements of MIL-STD-740-1 for grade A12 equipment.

3.5.4.2 Structureborne noise. Monitoring system structureborne noise levels shall not exceed the requirements of MIL-STD-740-2 for type III equipment.

3.5.5 Inclination. The monitoring system shall operate within the specified performance limits when subjected to attitudes and motions encompassed by DOD-STD-1399, Section 301A (see 4.7.34).

3.5.6 Contamination resistance.

3.5.6.1 Contaminating substances. The monitoring system shall operate (within specified accuracy) without damage when exposed to atmospheric constituents in the ranges indicated below. (Note, atmospheric constituents may be present in any combination.)

Stibine	0 to 1.0 ppm
Chlorine	0 to 3.0 ppm
Carbon monoxide	0 to 200 ppm
Carbon dioxide	0 to 4.0 percent
Sulphur dioxide	0 to 10.0 ppm
Dichlorodifluoromethane (R-12)	0 to 2,000 ppm

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Dichlorotetrafluoroethane (R-114)	0 to 2,000 ppm
Chlorotetrafluoroethane (R-124)	0 to 2,000 ppm
Tetrafluoroethane (R-134A)	0 to 2,000 ppm

3.5.6.2 Filters. Filters, where applicable, shall be incorporated in the monitoring system to negate the effect of any contaminating substances specified in 3.5.6.1 that may be in the atmosphere and would interfere with system performance. Filters shall be provided with a means for being readily inserted and replaced. Filter cartridges shall be furnished to provide for 5,000 hours of continuous operation when subjected to velocity conditions of up to 3,000 fpm.

3.6 Interchangeability. Parts having the same part number shall possess such functional, dimensional, and physical characteristics as to be equivalent in performance, reliability, maintainability, and durability. Parts shall be interchangeable (exchangeable one for the other) without alteration of the parts themselves or of adjoining parts, except for adjustment or recalibration, and without selection for fit or performance.

3.7 Safety. The monitoring system shall be such that the exposure of either of the sensors to a hydrogen-air atmosphere of up to 8 percent (0-8% H_2), by volume, shall not result in flashback or explosion, and shall not require the renewal of the flashback arrestors, if used, throughout the safety test of 4.7.37.

3.8 Drawings. (See 6.3).

3.9 Human engineering. The monitoring system shall be in conformance with the human engineering design criteria and requirements specified in ASTM F 1166 and MIL-H-46855, respectively.

3.10 Workmanship. The workmanship of the monitoring system shall be in accordance with requirement 9 of MIL-STD-454, and soldering shall conform to MIL-STD-2000.

3.11 Workmanship screen. Monitoring systems shall withstand a defect detection screen of random type vibration and temperature cycling as specified in 4.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall

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not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, inspections shall be made at the following conditions:

Temperature:	32° to 122°F (0° to 50°C)
Atmospheric pressure:	24 to 36 in-Hg
Relative humidity:	20 to 120 grains of water per pound of dry air

4.3.1 Controlled inspection conditions. When controlled environmental (temperature, pressure, and humidity) and velocity conditions are specified, the following conditions shall be maintained, as applicable:

Temperature:	77° ± 5°F (25° ± 2.8°C)
Atmospheric pressure:	30 in-Hg ± 0.5 in-Hg
Relative humidity:	70 ± 5 grains of water per pound of dry air
Velocity:	2,040 fpm ± 100 fpm

4.4 First article inspection. First article inspection shall be performed on the first unit produced. First article inspection shall also be performed when the basic design of, or material used in, the monitoring system is changed and such changes may affect the monitoring system's ability to comply with this specification. First article inspection shall consist of the tests specified in table I. Tests shall be performed in the order listed.

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TABLE I. First article inspection.

	Requirement	Inspection method
Workmanship screen	3.11	4.8 thru 4.8.2
Inspection of packaging	5 thru 5.2	4.6.3
General examination	3.2 thru 3.2.3, 3.3, 3.3.2.11, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.5, 3.3.4 thru 3.3.4.4 and 3.3.4.6 thru 3.3.4.9, 3.3.5 thru 3.3.5.2, 3.3.6 thru 3.3.6.4, 3.3.7, 3.3.8 thru 3.3.8.3, 3.3.9.1, 3.3.9.2, 3.4.2, 3.4.3, 3.4.6, 3.5.6.2, 3.9, 3.10	4.7.1
Power requirements	3.3.1	4.7.2
Insulation resistance	3.4.14	4.7.27
Burn-in test	3.4.13	4.7.26
Vibration	3.5.3	4.7.32
Shock	3.5.2	4.7.31
Inclination	3.5.5	4.7.34
Electromagnetic interference	3.4.4	4.7.25
Accuracy	3.3.2.2	4.7.3.4
Calibration check:		
Zero adjustment	3.3.9.1	4.7.20
Span adjustment	3.3.9.2	4.7.21
Range of measurement	3.3.2.1	4.7.3.3
Detection level	3.3.2.3	4.7.4
Repeatability/reproducibility	3.3.2.8	4.7.9
Response time	3.3.2.9	4.7.10
Recovery time	3.3.2.10	4.7.11
Controller output signal	3.3.4.5	4.7.16
Safety	3.7	4.7.37
Contamination resistance	3.5.6.1, 3.5.6.2	4.7.35
Accelerated life	3.3.2.12	4.7.12
Drift	3.3.2.13, 3.3.2.14	4.7.13
Electrical stability	3.5.1.1	4.7.28
Temperature	3.3.2.4	4.7.5
Thermal stability	3.5.1.2	4.7.29
Pressure	3.3.2.5	4.7.6
Pressure stability	3.5.1.3	4.7.30
Humidity	3.3.2.6	4.7.7
Velocity effects	3.3.2.7	4.7.8
Pressure loss	3.3.3.6	4.7.14
Leakage	3.3.3.7	4.7.15
Noise	3.5.4.1, 3.5.4.2	4.7.33
Splash test	3.3.4.6, 3.3.5.1	4.7.17
Status indication	3.3.8, 3.3.8.1, 3.3.8.2	4.7.18
Reset	3.3.8.3	4.7.19

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TABLE I. First article inspection. - Continued

	Requirement	Inspection method
Self-diagnostics	3.4.1	4.7.22
Time delay	3.4.2	4.7.23
Test mode	3.4.3	4.7.24
Interchangeability	3.6	4.7.36

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of products of the same type, produced under identical conditions, and offered for delivery at the same time.

4.5.1.2 Group A inspection. Each monitoring system shall be subjected to group A inspection. Group A inspection shall consist of the tests specified in table II. Tests shall be performed in the order listed and the results of each test compared with this specification. Failure to conform to this specification for a group A inspection test shall be cause for the monitoring system to be rejected. If a monitoring system is rejected, the contractor may rework it to correct the defects and resubmit it for reinspection.

TABLE II. Quality conformance inspection.

	Requirement	Inspection method
Group A		
Workmanship screen	3.11	4.8 thru 4.8.2
General examination	3.2 thru 3.2.3, 3.3, 3.3.2.11, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.5, 3.3.4 thru 3.3.4.4 and 3.3.4.6 thru 3.3.4.9, 3.3.5 thru 3.3.5.2, 3.3.6 yhtu 3.3.6.2, 3.3.7, 3.3.8 thru 3.3.8.3, 3.3.9.1, 3.3.9.2, 3.4.2, 3.4.3, 3.4.6, 3.5.6.2, 3.8, 3.10	4.7.1
Insulation resistance	3.4.14	4.7.27
Burn-in test	3.4.13	4.7.26
Accuracy	3.3.2.2	4.7.3.4

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TABLE II. Quality conformance inspection. - Continued

	Requirement	Inspection method
Calibration check:		
Zero adjustment	3.3.9.1	4.7.20
Span adjustment	3.3.9.2	4.7.21
Range of measurement	3.3.2.1	4.7.3.3
Detection level	3.3.2.3	4.7.4
Repeatability/reproducibility	3.3.2.8	4.7.9
Response time	3.3.2.9	4.7.10
Recovery time	3.3.2.10	4.7.11
Controller output signal	3.3.4.5	4.7.16
Group B		
Status indication	3.3.8, 3.3.8.1, 3.3.8.2	4.7.18
Reset	3.3.8.3	4.7.19
Self-diagnostics	3.4.1	4.7.22
Time delay	3.4.2	4.7.23
Test mode	3.4.3	4.7.24

4.5.1.3 Group B inspection. Monitoring systems shall be subjected to group B inspection. Group B inspection shall consist of the tests specified in table II. Tests shall be performed in the order listed and the results of each test compared with this specification. As a minimum, the contractor shall inspect a sample of monitoring systems in accordance with table III. If one or more defects is found in any sample, the entire lot shall be rejected and screened 100 percent by the contractor for the defect found.

TABLE III. Sampling for group B inspection.

Number of monitoring systems in an inspection lot	Number of monitoring systems in sample
3 and under	All
4 to 8	3
9 to 15	4
16 to 25	5
26 to 40	6
41 to 65	8
66 to 110	10
111 to 180	13
181 to 300	17
301 and over	22

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4.6 Inspection of packaging. Sample packs and the inspection of the preservation, packaging, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

4.7 Methods of inspection.

4.7.1 General examination. The monitoring system shall be given an examination to determine that it conforms to applicable specifications and approved working drawings with respect to material, finish, workmanship, construction, assembly, dimensions, weight and marking of identification and information plates. During examination, the monitoring system shall not be disassembled in such a way that its performance, durability, or appearance will be affected. This examination shall include a check of the operating controls, circuit functions, and adjustments, as applicable, and shall verify compliance with the requirements identified in table I for first article inspection and table II for quality conformance inspection.

4.7.2 Power requirements. The monitoring system shall be connected in the circuitry to verify that the input voltage and frequency are as specified in 3.3.1. Power requirements shall be evaluated under steady-state conditions. Evidence that the input voltage or frequency is not in accordance with the specified values (see 3.3.1) shall be cause for rejection.

4.7.3 Range and accuracy.

4.7.3.1 Calibration gases. Certified standard calibration gases shall be utilized for accuracy determinations, for zero, intermediate, and span adjustments, and for system evaluations. Calibration gases shall be certified zero air, 0.1 to 4.0 percent hydrogen-air (0.1-4.0% H_2), 8.0 percent hydrogen-air (8.0% H_2), and 0.1 to 4.0 percent hydrogen-nitrogen (0.1 - 4.0 H_2) mixtures, as applicable. Calibration gases shall be certified to be within plus or minus 0.02 percent absolute ($\pm$ 0.02% H_2), or plus or minus 2.0 percent ($\pm$ 2.0%) of the minor component, whichever is smaller

4.7.3.2 Consistent use of calibration gases. When hydrogen-air mixtures are used to calibrate the monitoring system, hydrogen-air mixtures shall also be used for accuracy determinations, accuracy verifications, and system evaluations. Similarly, when hydrogen-nitrogen mixtures are used to calibrate the monitoring system, hydrogen-nitrogen mixtures shall also be used for accuracy determinations, accuracy verifications, and system evaluations.

4.7.3.3 Range of measurement. Compliance with 3.3.2.1 shall be established based on the results of the accuracy test and span adjustment test specified, respectively, in 4.7.3.4 and 4.7.21.

4.7.3.4 Accuracy. The accuracy of the monitoring system shall be determined by exposing each sensor to certified zero air, hydrogen-air, and hydrogen-nitrogen calibration gases (see 4.7.3.1), as applicable, of 0, 1.0, 1.5, 2.0, 3.0, and 4.0 percent (0, 1.0, 1.5, 2.0, 3.0, and 4.0% H_2). Upscale and downscale determinations shall be performed. During each determination, the sensors shall be exposed to each specified calibration gas for a minimum of 5 minutes. The readout accuracy of the monitoring system during these determinations shall be as specified in 3.3.2.2. Prior to making an initial

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determination, the system shall be calibrated and each sensor shall be exposed to a certified zero air for a minimum of 5 minutes at approximately 25°C (77°F). During the determinations, the monitoring system shall be subjected to environmental conditions specified in 4.3.1. Accuracy determinations shall be made prior to each subsequent first article inspection test (see table I), prior to each group A inspection, and prior to group B inspection.

4.7.4 Detection level. The monitoring system detection level shall be determined by exposing each sensor to hydrogen-air or hydrogen-nitrogen mixtures, as applicable, and noting the minimum concentration at which a response is observed. During the determinations, the monitoring system shall be subjected to environmental conditions specified in 4.3.1. A minimum detection level less than that specified in 3.3.2.3 shall be cause for rejection. The readout accuracy at each determination shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4.

4.7.5 Temperature. Compliance with the requirements of 3.3.2.4 shall be demonstrated by verifying that the operational performance of the monitoring system during and after subjections to constant exposure at each specified temperature extreme, plus or minus 2°C (± 3.5°F), for a period of 4 hours is comparable to operational performance at ambient conditions. During the test the monitoring system shall be energized, positioned in its normal operating configuration, and subjected to pressure and humidity conditions specified in 4.3.1. In addition, each sensor shall be intermittently exposed to a 1.5 percent hydrogen-air or hydrogen-nitrogen mixture (1.5% H_2), as applicable, at least once per hour for a minimum of 5 minutes. The readout accuracy prior to and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. The readout accuracy during the test shall be as specified in 3.3.2.2.

4.7.6 Pressure. Compliance with the requirements of 3.3.2.5 shall be demonstrated by verifying that the operational performance of the monitoring system during and after subjection to constant exposure at each specified pressure limit, plus or minus 0.5 inches of mercury (± 0.5 in-Hg) for a period of 2 hours is comparable to operational performance at ambient conditions. During the test, the monitoring system shall be energized, positioned in its normal operating configuration, and subjected to temperature and humidity conditions specified in 4.3.1. In addition, each sensor shall be intermittently exposed to a 1.5 percent hydrogen-air or hydrogen-nitrogen mixture (1.5% H_2), as applicable, at least once per hour for a minimum of 5 minutes. The readout accuracy during and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. The readout accuracy during the test shall be as specified in 3.3.2.2.

4.7.7 Humidity. The monitoring system shall be subjected to and pass a humidity test in accordance with method 507.2, procedure III, of MIL-STD-810. During the test, the monitoring system shall be energized, positioned in its normal operating configuration, and subjected to pressure conditions specified in 4.3.1. In addition, each sensor shall be intermittently exposed to a 1.5 percent hydrogen-air or hydrogen-nitrogen mixture (1.5% H_2), as applicable, at least once per hour for a minimum of 5 minutes. The readout accuracy prior to and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. The readout accuracy during the test shall be as specified in 3.3.2.2.

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4.7.8 Velocity effects. Compliance with the requirements of 3.3.2.7 shall be demonstrated by verifying that the readout accuracy of the monitoring systems when the sensors are mounted in a 12-inch diameter (12" I.D.) circular ventilation duct and subjected to constant exposure at each specified velocity extreme plus or minus 5.0 percent (± 5.0%) for a period of 1 hour, is as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. During the test, the sensors shall be intermittently exposed to a 1.5 percent hydrogen-air or hydrogen-nitrogen mixture (1.5% H_2), as applicable, at least 4 times per hour for a minimum of 5 minutes. Exposures shall be made at 15-minute intervals. The readout accuracy during the test shall be as specified in 3.3.2.2.

4.7.9 Repeatability/reproducibility. Monitoring system repeatability/reproducibility shall be determined based on a series of tests conducted in accordance with 4.7.3.4. The series shall consist of a minimum of three trials. Upscale and downscale determinations shall be performed to evaluate repeatability/reproducibility. Repeatability/reproducibility shall be as specified in 3.3.2.8.

4.7.10 Response time. Monitoring system response time shall be evaluated based on the exposure of each sensor to both a step input of a hydrogen-air or hydrogen-nitrogen mixture of 1.5 percent (1.5% H_2), as applicable, and a hydrogen-air or hydrogen-nitrogen mixture of 4.0 percent (4.0% H_2), as applicable. During the test, the sensors and remote indicators shall be located a minimum of 50 feet and 350 feet, respectively, from the controller. The readout accuracy prior to and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. The readout accuracy during the test shall be as specified in 3.3.2.2. The response time shall be not more than that specified in 3.3.2.9.

4.7.11 Recovery time. Monitoring system recovery time shall be determined based on the time required for the system to return to its normal ambient reading following the removal of applied hydrogen-air and hydrogen-nitrogen mixtures during the response time test specified in 4.7.10. The recovery time shall be not more than that specified in 3.3.2.10.

4.7.12 Accelerated life. Compliance with the requirements of 3.3.2.12 shall be demonstrated by verifying that subjecting the monitoring system to an accelerated life test in accordance with MIL-E-16400 does not cause damage or operational performance degradation. Prior to initiating the test, the system shall be calibrated and steady-state conditions shall be established. During the test, the system shall be intermittently exposed to a hydrogen-air or hydrogen-nitrogen mixture of 1.5 percent (1.5% H_2), as applicable, at least twice per 24 hours for a minimum of 5 minutes. Exposures shall be made at 12-hour intervals. The readout accuracy prior to and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. The readout accuracy during the test shall be as specified in 3.3.2.2.

4.7.13 Drift. During and after the accelerated life test specified in 4.7.12, the controller and remote indicator meters shall be checked for zero and span calibration drift. Zero drift shall be as specified in 3.3.2.13. Span calibration drift shall be as specified in 3.3.2.14.

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4.7.14 Pressure loss. Compliance with 3.3.3.6 shall be demonstrated by verifying that the pressure loss across the sensors, when mounted in a 12-inch diameter (12" I.D.) circular ventilation duct and subjected to velocity conditions specified in 4.3.1, is not greater than 0.1 inch of water (0.1 in w.g.).

4.7.15 Leakage. Compliance with 3.3.3.7 shall be demonstrated by mounting the sensor in a 12-inch diameter (12" I.D.) circular ventilation duct and verifying that no evidence of air, gas or moisture transmission is observed between the sensor enclosure and duct when the duct is subjected to internal pressures of up to 2.0 lb/in^2g.

4.7.16 Controller output signal. After zero and span adjustment (see 4.7.20 and 4.7.21, respectively), an output signal corresponding to zero shall be indicated at each controller and remote indicator meter when the respective sensor is exposed to a zero air. Similarly, an output signal corresponding to the full scale value shall be indicated at each controller and remote indicator meter when the respective sensor is exposed to a hydrogen-air or hydrogen-nitrogen mixture, as applicable, equal to the upper limit of the specified range of measurement (see 3.3.2.1). During the tests, the sensors and remote indicators shall be located a minimum of 50 feet and 350 feet, respectively, from the controller. Controller output signal tolerance shall be as specified in 3.3.4.5. A minimum of three determinations shall be made for each sensor. The readout accuracy during the test shall be as specified in 3.3.2.2. The readout accuracy prior to and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4.

4.7.17 Splash test. The controller and remote indicator enclosures shall be subjected to and pass the splashproof test specified in MIL-STD-108. The readout accuracy after completion of this test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4.

4.7.18 Status indication. Compliance with the requirements of 3.3.8.1 shall be demonstrated by exposing each sensor to increasing hydrogen-air mixtures ranging from 0.5 to 2.5 percent (0.5-2.5% H_2) in 0.5 percent increments (0.5% H_2) and verifying that the alarm activation level is as specified. Upon reaching a 2.5 percent hydrogen-air mixture (2.5% H_2), the hydrogen concentration shall be gradually reduced to 0.5 percent (0.5% H_2) in 0.5 percent increments (0.5% H_2). A minimum of three determinations shall be made for each sensor. The readout accuracy after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. Compliance with the requirements of 3.3.8.2 shall be demonstrated by isolating and simulating (individually and collectively) each of the specified malfunction conditions and verifying that audible and visual malfunction indicators and contacts are properly activated.

4.7.19 Reset. Compliance with the requirements of 3.3.8.3 shall be demonstrated during the test specified in 4.7.18 by energizing the manual reset and verifying that applicable audible and visual indicators and contacts are deactivated as specified in 3.3.8.3.

4.7.20 Zero adjustment. Compliance with 3.3.9.1 shall be demonstrated by verifying that the exposure of each sensor to zero air results in respective controller and remote indicator meter readings within the accuracy

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specified in 3.3.2.2. A minimum of three determinations shall be made for each sensor. Prior to initiating the test, steady-state conditions shall be established. During each determination, the sensors shall be mounted in a 12-inch diameter (12" I.D.) circular ventilation duct and subjected to a velocity of 315 fpm plus or minus 5.0 percent (± 5.0%).

4.7.21 Span adjustment. Compliance with 3.3.9.2 shall be demonstrated by verifying that the exposure of each sensor, after zero adjustment (see 4.7.20), to a hydrogen-air or hydrogen-nitrogen mixture, as applicable, equal to the upper limit of the specified range of measurement (see 3.3.2.1) results in respective controller and remote indicator meter readings within the accuracy specified in 3.3.2.2. A minimum of three determinations shall be made for each sensor. Prior to initiating the test, steady-state conditions shall be established. During each determination the sensors shall be mounted in a 12-inch diameter (12" I.D.) circular ventilation duct and subjected to a velocity of 315 fpm plus or minus 5.0 percent (± 5.0%).

4.7.22 Self-diagnostics. Compliance with the requirements of 3.4.1 shall be demonstrated by verifying that specified self-diagnostic routines have been provided, continuity exists within each specified circuit, and simulated or induced faults within each specified circuit are identified.

4.7.23 Time delay. Compliance with the requirements of 3.4.2 shall be demonstrated by engaging the time delay and verifying that alarm and malfunction indicator contacts are not activated prior to the expiration of the time delay when the monitoring system is subjected to induced false indications as specified in 3.4.2.

4.7.24 Test mode. Compliance with the requirements of 3.4.3 shall be demonstrated by engaging the test mode while the monitoring system is in both an active and passive mode and verifying that the system functional status and calibration are indicated, and that the alarm and malfunction contacts or circuits are disabled.

4.7.25 Electromagnetic interference. Compliance with the requirements of 3.4.4 shall be demonstrated by performing and passing electromagnetic emission and susceptibility tests in accordance with MIL-STD-462. The results of the electromagnetic emission and susceptibility tests, and visual inspection are to be documented in an EMI test report (see 6.3).

4.7.26 Burn-in test. The monitoring system shall be subjected to 24 hours of continuous operation. Following completion of the 24-hour period, the readout accuracy shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. Prior to initiating the test, the system shall be calibrated and steady-state conditions shall be established. During the test, the system shall be intermittently exposed to a hydrogen-air or hydrogen-nitrogen mixture of 1.5 percent (1.5% H_2), as applicable, at least once per hour for a minimum of 5 minutes. Exposures shall be made at 1 hour intervals. The readout accuracy during the test shall be as specified in 3.3.2.2.

4.7.27 Insulation resistance. Compliance with the requirements of 3.4.14 shall be demonstrated by verifying that the insulation resistance of the monitoring system electrical circuits is not less than 10 megohms at

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specified environmental service conditions (see 4.3) when subjected to 500 Vdc superimposed on the ac power supply.

4.7.28 Electrical stability. The monitoring system shall be subjected to and pass voltage and frequency tolerance tests, voltage and frequency transient tolerance and recovery tests, and voltage spike tests in accordance with MIL-STD-1399, Section 300A. Prior to initiating the test, the system shall be calibrated and steady-state conditions shall be established. During the test, the monitoring system shall be subjected to environmental conditions specified in 4.3.1. Testing shall not cause damage to the system (except for protective fuses). The readout accuracy prior to and after the test shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4.

4.7.29 Thermal stability. Thermal stability of the monitoring system shall be established by subjecting the system to a 24-hour cycling sequence. The monitoring system shall be energized and operated during this test. Prior to initiating the test, the system shall be calibrated and steady-state conditions shall be established. During the test, the monitoring system shall be subjected to pressure and humidity conditions specified in 4.3.1. After establishing steady-state conditions, the temperature shall be cycled between the upper and lower temperature limits corresponding to transient conditions specified in 3.5.1.2. The rate of temperature change shall not exceed 10°C (18°F) per minute. Cycle transitions shall be as smooth as practical. The exposure time at each limit shall be 2 hours. At the end of the 24-hour period steady-state conditions shall be re-established. The readout accuracy during the last cycle and after steady-state conditions have been re-established shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. Accuracy measurements taken during the last cycle shall be made 1 hour prior to the completion of each segment of the cycle. Evidence of a system malfunction during this test shall also constitute a failure.

4.7.30 Pressure stability. Pressure stability of the monitoring system shall be established by subjecting the system to two 12-hour cycling sequences. The monitoring system shall be energized and operated during this test. Prior to initiating the test, the system shall be calibrated and steady-state conditions shall be established. During the test, the monitoring system shall be subjected to temperature and humidity conditions specified in 4.3.1. After establishing steady-state conditions, the pressure shall be cycled between the upper and lower pressure limits corresponding to conditions specified in 3.5.1.3. The rate of pressure change shall not exceed 2.7 in-Hg per minute. Cycle transitions shall be as smooth as practical. During the first 12-hour cycling sequence each cycle shall be completed within 3 minutes During the second 12-hour cycling sequence, the exposure time at each limit shall be 1 hour. Steady-state conditions shall be re-established at the end of each 12-hour period. The readout accuracy after steady-state conditions have been re-established and during the last cycle of each 12-hour cycling sequence shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. Accuracy measurements taken during the last cycle of each 12-hour cycling sequence shall be made 15 minutes prior to the completion of each segment of the cycle. Evidence of a system malfunction during the test shall also constitute a failure.

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4.7.31 Shock. The monitoring system shall be subjected to and pass a grade A, class I, type A high impact shock test in accordance with MIL-S-901. The monitoring system shall be energized during the test. After the test, the monitoring system shall be subjected to and pass inspections as specified in 4.7.3.4, 4.7.10, and 4.7.11.

4.7.32 Vibration. The monitoring system shall be subjected to and pass a type I environmental vibration test in accordance with MIL-STD-167-1. Exciting frequency ranges up to and including 25 hertz (Hz) shall be evaluated during the vibration test. The monitoring system shall be energized during the test. After the test, the monitoring system shall be subjected to and pass inspection as specified in 4.7.3.4, 4.7.10, and 4.7.11.

4.7.33 Noise. The monitoring system shall be tested in accordance with MIL-STD-740-1 to determine conformance to 3.5.4.1. The monitoring system shall also be tested in accordance with MIL-STD-740-2 to determine conformance to 3.5.4.2.

4.7.34 Inclination. Compliance with the requirements of 3.5.5 shall be demonstrated by verifying that the accuracy of the monitoring system remains within the specified limits during and after subjection to an inclination test. The monitoring system shall be tilted 30 degrees from vertical and operated for 15 minutes in each of four directions with the sensors exposed to a 1.5 percent hydrogen air or hydrogen-nitrogen mixture (1.5% H_2).

4.7.35 Contamination resistance. The contamination resistance of the monitoring system to specified atmospheric constituents (see 3.5.6.1) shall be collectively and individually evaluated during a series of 1-hour contamination resistance tests. The collective evaluation shall be made by intermittently exposing each sensor for 5 minutes to a 1.5 percent hydrogen-air or hydrogen-nitrogen mixture (1.5% H_2), as applicable, containing eight of the specified atmospheric constituents (stibine is excluded). Exposures shall be made twice per hour at 30-minute intervals. Individual evaluations shall be made by intermittently exposing each sensor for 5 minutes to 1.5 percent hydrogen-air or hydrogen-nitrogen mixtures (1.5% H_2), as applicable, containing one specified atmospheric constituent (stibine is excluded). Exposures shall be made twice per hour at 30-minute intervals. The concentration of the atmospheric constituents for each evaluation shall be equal to the maximum concentration specified in 3.5.6.1. The monitoring system shall have steady-state conditions established. During the tests, the monitoring system shall be subjected to environmental conditions specified in 4.3.1. The readout accuracy during and after each evaluation shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4. The readout accuracy during each evaluation shall be as specified in 3.3.2.2.

4.7.36 Interchangeability. Compliance with the requirements of 3.6 shall be demonstrated through the ability to remove and substitute each of the normally replaceable parts with a part having the same part number. For purposes of this test, normally replaceable parts shall constitute each sensor, each controller and remote indicator meter, and each module. An exchange shall represent the concurrent removal and substitution of each normally replaced part. Two successful exchanges without duplication shall be

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made for each item. After each exchange, the monitoring system shall be recalibrated. The readout accuracy after each exchange shall be as specified in 3.3.2.2 when verified in accordance with 4.7.3.4.

4.7.37 Safety. The monitoring system shall be energized and tested for safety against flashback or explosion by exposing each sensor to hydrogen-air mixtures of 8 percent (8% H_2). A minimum exposure period of 1 minute shall be utilized for this test. Evidence of flashback or explosion shall constitute a failure.

4.8 Stress screen program. The contractor shall develop and implement a program for stress screening of electronic hardware. The objective of the program is to expose electronic hardware to electrical and environmental stress screening at the part level, and one or more levels of assembly on a 100 percent basis. The program shall include screening of procedures at the part level (vendor and incoming inspection) as well as module, unit, and system level, (see 6.3). The types or manner of application, and level of severity of each stress screen shall be defined. The plan shall be approved by NAVSEA prior to program implementation.

4.8.1 Manufacturing stress screens. Manufacturing stress screens should be designed and proofed on specific hardware to determine the most effective screening environments and levels and at the same time determine the environments and levels that would cause system degradation due to overstress. Unless more effective screens are determined through such proofing experiments, the stress screening program shall include environments and levels as follows:

Thermal Cycling:

Temperature range:	minus 40 to plus 95 degrees Celsius
Temperature rate of change:	5 degrees celsius per minute minimum as measured by thermocouples mounted directly on electronic components
Thermal cycles:	10 to 20 cycles as a minimum depending on level of assembly (last cycle failure free)
Operating conditions:	Power "OFF"

Random Vibrations:

Duration:	10 minutes per axis
Number of axes:	2 axes
Acceleration spectrum:	0.04 G^2/Hz from 20 to 2000 Hz with a 3dB/octave rolloff from 80 to 20 Hz and 350 to 2000 Hz (where G is the acceleration due to gravity and dB is decibels).
Operating conditions:	Power "OFF"

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4.8.2 Part screening. The stress screening program shall include part screening at the equipment contractor's facility to prevent introduction of defective parts in the manufacturing process. The following tests shall be performed as a minimum:

(a) Electrical test at +25° and +125°C (Static, Dynamic and Functional)
(b) Particle Impact Noise Test (PINT) as a minimum on all hybrids and all unglassivated semiconductors with cavities
(c) Destructive physical analysis (minimum 2 parts per lot).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.6)

5.1 Navy Fire-Retardant Requirements

a. Treated Lumber and Plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

b. Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of interior (unit and intermediate) and exterior fiberboard boxes including interior packaging forms shall conform to the class-domestic/fire retardant or class-weather resistant/fire retardant materials requirements as specified (see 6.2) of PPP-F-320 and amendments thereto.

5.2 Packaging requirements. The packaging (preservation, packing and marking) requirements shall be in accordance with MIL-E-17555 for the level of preservation, the level of packing, including marking and packaging acquisitioning option therein, as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hydrogen monitoring system is intended for use on submarines for continuously and simultaneously monitoring and indicating the concentration of hydrogen in both forward and aft battery compartment exhaust ductwork.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. First article sample, if required (see 3.1).
d. Duct type and size (see 3.3.2.7, 3.3.3.2, 3.3.3.6, and 3.3.3.7).
e. Transmission distance of intermediate signal amplifiers, if required (see 3.4.9).
f. Navy fire-retardant material requirements (see 5.1).
g. Preservation, packing, and marking requirements (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID number	DID title	Tailoring
3.8	DI-DRPR-81000	Product drawings, and associated lists	Provide only level 3
4.7.25	DI-EMCS-80200	Electromagnetic interference test report	--
4.7.31	DI-ENVR-80708	Reports, equipment shock test	--
4.7.32	DI-ENVR-80706	Report, vibration testing	--
4.7.33	DI-HFAC-80272	Equipment airborne sound measurements test report	--
4.7.33	DI-HFAC-80274	Equipment structureborne vibration acceleration measurements test report	--
4.8	DI-R-7079	Reliability program plan	--
4.8	DI-MGMT-80909	Program plan	--

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data

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Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot production items, a standard production item from the contractors current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and dispostion of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4.1 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (keyword) listing.

Battery compartment
Detector
Exhaust ductwork
Gas detector
Indicating
Indicator
Sensor

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6625-N906)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-2220C(SH)	2. DOCUMENT DATE (YYMMDD) 1991/January/16

3. DOCUMENT TITLE: INSTRUMENTS, ELECTRICAL-MEASURING, HYDROGEN-GAS-MONITORING, NONPORTABLE (SUBMARINE USE)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TOPC): Mr. Peter Schilling (SEA 56Y13) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: TPOC: 703-602-5473 (2) AUTOVON: 8-332-5473
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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